For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GENERAL CERTIFICATE OF EDUCATION ADVANCED LEVEL

GEOGRAPHY
PAPER 3 Practical

6037/3
3 hours

SPECIMEN PAPER

2 x 1: 50 000 survey maps of Zimbabwe are enclosed with this question paper.

Additional materials:
Answer booklet
Mathematical formula booklet
Calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

Write your answers on the separate answer booklet provided.

Answer **three** questions, one question from each of the Sections **A**, **B** and **C**.

If you use more than one booklet, fasten them together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

Sketch maps and diagrams should be drawn whenever they serve to illustrate an answer.

This paper consists of **9** questions.

This specimen paper consists of 8 printed pages.

SECTION A

STATISTICS

Answer **one** question only.

1 **(a)** Define the following statistical terms:

(i) Mode, [2]

(ii) Median, [2]

(iii) Range. [2]

(b) **Table 1.1** shows the height of 10 stalks of a variety of sorghum plants.

Table 1.1

stalk height in (cm)	140	145	150	155	160	165	170	175	180	185
Frequency	3	2	5	7	7	9	10	8	7	5

(i) Use the data in **Table 1.1** to find the range, mode and mean. [3]

(ii) Draw a frequency polygon for the data set in **Table 1.1**. [5]

(c) A school claims that the average reading score of its students is at least 70. Given that n = 16, $\overline{x} = 68$ and s = 9.

(i) Test at 5% level of significance that M is 70. [9]

(ii) Comment on the school's claim. [1]

(iii) What is the implication of committing type 1 error in hypothesis testing? [1]

2 **(a)** Distinguish between bivariate and univariate statistics. [2]

(b) **Table 2.1** shows area under cultivation for different crops in one resettlement area in Zimbabwe.

Table 2.1

crop	Tobacco	Maize	Vegetable	Fruits	Wheat	Barley	Pasture
Hectares	10	12	15	3	29	18	17

(i) State **one** advantage of the use of mode in the interpretation of statistical data. [1]

(ii) Use the data in **Table 2.1** to calculate the interquartile range. [4]

(c) **Table 2.2** shows daily income of a bottle store for a period of 7 days.

DAY	Sun	Mon	Tue	Wed	Thur	Fri	Sat
INCOME	$240	$220	$240	$300	$370	$570	$590

(i) Use data in **Table 2.2** to construct a line graph. [5]

(d) **Table 2.3** shows percentages on unemployment and mortality rate in Southern African Development Community (SADC).

Table 2.3

Country	Malawi	Botswana	South Africa	Zambia	Tanzania	Swaziland	Lesotho	Zimbabwe	Angola
unemployment	9	6	6	7	8	12	10	24	17
Mortality	9	6	5	8	9	14	11	15	13

(i) Use the data in **Table 2.3** to calculate Pearson's correlation coefficient. [10]

(ii) Comment on the relationship between unemployment and mortality in SADC countries. [3]

3 (a) (i) Define the terms, population and sample. [4]

(ii) **Table 3.1** shows the number of people who arrive at a clinic between 1 pm and 3 pm.

Table 3.1

Day	1	2	3	4	5
Number of people	4	8	5	2	1

Use the data in **Table 3.1** to calculate the standard deviation. [5]

(b) **Table 3.2** shows tourist destination preferences in Zimbabwe.

Table 3.2

Destination Centre	**Number of tourist destination preferences**
Chinhoyi Caves	30
Kariba	40
Victoria Falls	50
Matopos National Park	20
Gonarezhou National Park	20

Draw a pie chart to show preferences on tourist destinations in **Table 3.2** [6]

(c) **Table 3.3** below shows the number of advertisements placed in a local newspaper and the revenue generated.

Table 3.3

number of advertisements	10	3	6	2	4
revenue generated ($)	400	150	250	100	300

(i) Use the data in **Table 3.3** to draw a scatter graph. [5]

(ii) Find the linear regression equation of the revenue generated and number of advertisements.
Comment on the results. [5]

SECTION B

MAPPING AND GEOGRAPHIC INFORMATION SYSTEMS

Answer **one** question only.

4 **(a)** Study the 1:50 000 map of Rusape, Zimbabwe.

Draw an annotated sketch section to show variations in relief between grid point 4001007936000 and 4090007936000. [6]

(b) Explain the relief and drainage features shown on the map. [12]

(c) Explain how drainage has influenced people's activities in Rusape as shown on the map. [7]

5 **(a)** You are required to perform geographical data input, processing and output procedures in QGIS. You are provided with a scanned map of Rusape

(i) Open QGIS and add the following coordinates in georeferencer. Produce a hard copy.

EASTINGS	NORTHINGS
406000	7940000
412000	7940000
412000	7934000
406000	7934000

[4]

(ii) Georeference the map and produce a hard copy. You can use Arc 1950 zone 36S CRS. [2]

(b) Use the georeferenced map to digitize the following features on the map:

(i) Nyazura built up area. [3]

(ii) Nyamatsanga river and label it. [2]

(iii) the road from grid point 4120007930000 to where it crosses Northing 7939000. [2]

(iv) Tikwiri hill and label it. [2]

(v) Any **three** dip tanks and label them. [3]

[Turn over

(vi) Use QGIS software to show the area served by each dip tank, if their sphere of influence's range is 1000m. Set transparency to 50%.

Produce a hard copy. [3]

(c) (i) Measure the length of the aerodrome in Nyazura. [2]

(ii) Measure the area of a plantation west of Nyazura. [2]

6 (a) (i) Explain the meaning of electro-magnetic spectrum. [4]

(ii) What does BGR stand for in remote sensing? [2]

(iii) Describe the following ways of image acquisition:
Across track scanning,
Along track scanning.

[6]

(b) **Figure 6** below shows spectral reflectance curve of a healthy tomato.

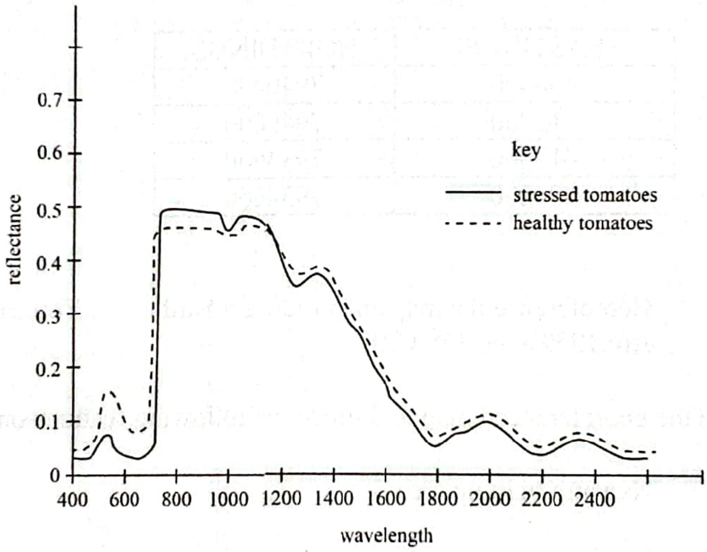

Fig.6

(i) Explain the term spectral reflectance signature. [3]

(ii) Explain the spectral reflectance curve of a healthy tomato shown in **Fig.6** [4]

(c) Comment on the implication of higher and low Normalised Vegetation Density Index (NVDI) to a forest manager. [6]

SECTION C

RESEARCH TECHNIQUES

Answer **one** question only.

7 **(a)** **Table 7** shows transpiration rates and rainfall totals recorded at a weather station.

Table 7

Month	J	F	M	A	M	J	J	A	S	O	N	D
Transpiration rates (mm/min)	30	28	26	22	18	12	10	18	24	28	30	32
Mean monthly rain fall(mm)	80	116	76	40	14	6	4	10	18	38	98	122

(i) What apparatus is used to measure transpiration rate? [2]

(ii) Describe how transpiration is measured. [6]

(iii) Use the information in the **Table 7** to construct transpiration-rainfall graph. [8]

(b) Explain the significance of evapotranspiration rates to farmers. [4]

(c) Explain the application of Indigenous Knowledge Systems (IKS) in weather forecast. [5]

8 **(a)** **(i)** With the aid of diagrams describe;
cluster sampling,
systematic sampling,
quadrant sampling. [3]

(ii) Describe how systematic sampling is used in any **one** demographic research. [4]

(b) **Table 8** shows distance travelled by a customer to a shopping centre.

Village of origin	A	B	C	D	E	F	G	H	I	J
Distance travelled (km)	6	7	9	12	3	15	8	11	10	5

(i) Use the information in **Table 8** to draw a desire line map to show the sphere of influence of the shopping centre. [5]

(ii) Explain the importance of a desire line map to a town planner. [3]

(iii) Describe any **one** other method that can be used to determine the sphere of influence of a shopping centre. [5]

(c) Explain how interviews can be used in collecting demographic data for economic development in Zimbabwe. [5]

9 (a) (i) What is a *pandemic*? [3]

(ii) Choose any **one** pandemic and describe how it spreads. [5]

(iii) Suggest practical measures that can be taken to prevent future outbreak of the named pandemic disease in Zimbabwe. [7]

(b) Traffic congestion has become a major problem in Zimbabwe's large cities.
Describe practical steps that are taken to arrive at this conclusion. [4]

(c) Show how the problem of traffic congestion can be solved in large cities such as Harare. [6]